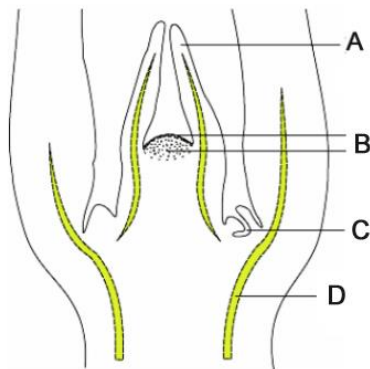


6. ANATOMY OF FLOWERING PLANTS

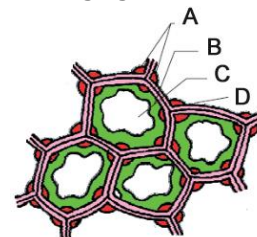
THE TISSUES

- A tissue is a group of cells having
 - Common origin and function
 - Common structure and function
 - Common origin and structure
 - All the above
- Plant tissues are classified into Meristematic and Permanent tissues based on the
 - Position of tissues
 - Capability of cell division
 - Shape of the cells
 - Number of types of cells
- The tissues where active cell division and growth occurs are called
 - Meristems
 - Intercalary meristems
 - Apical meristems
 - Lateral meristems
- Some cells 'left behind' from shoot apical meristem, constitute the
 - Dermal tissues
 - Secondary meristem
 - Axillary bud
 - Leaf primordium
- Label the figure correctly



- A= Shoot apical meristem, B= Leaf primordium, C= Axillary bud, D= Differentiating vascular tissue
 - A= Leaf primordium, B= Shoot apical meristem, C= Axillary bud, D= Differentiating vascular tissue
 - A= Axillary bud, B= Leaf primordium, C= Shoot apical meristem, D= Differentiating vascular tissue
 - A= Axillary bud, B= Shoot apical meristem, C= Leaf primordium, D= Differentiating vascular tissue
- Primary meristems are
 - Apical and intercalary meristems
 - Apical and lateral meristems
 - Lateral and intercalary meristems
 - Apical, intercalary and lateral meristems
 - Which meristem is occurred in regenerate parts removed by the grazing herbivores?
 - Apical meristem
 - Intercalary meristem
 - Lateral meristem
 - Secondary meristem
 - Intra fascicular cambium is an example of
 - Promeristem
 - Primary meristem

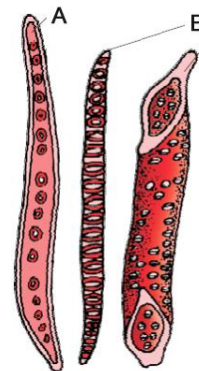
- Plate meristem
 - Secondary meristem
- Which of the following is the function of primary meristems?
 - They contribute to the formation of primary plant body.
 - Specific regions of the apical meristem produce dermal tissues and ground tissues.
 - They produce vascular tissues.
 - All of these
 - Fascicular vascular cambium, interfascicular cambium & cork cambium belong to
 - Primary meristem
 - Intercalary meristem
 - Secondary meristem
 - Apical meristem
 - Which one is false statement regarding lateral meristem?
 - They occur in mature regions of roots & shoots.
 - They are cylindrical meristems.
 - They are seen in ferns, gymnosperms and all angiosperms.
 - These are responsible for producing the secondary tissues.
 - The cells of permanent tissues are
 - Differentiated from meristematic cells
 - Having no ability to divide
 - Structurally and functionally specialized
 - All the above
 - Within a plant organ, the major tissue is
 - Parenchyma
 - Collenchyma
 - Sclerenchyma
 - Xylem
 - Select the false statement.
 - Simple permanent tissues include Parenchyma, Collenchyma and Sclerenchyma.
 - In collenchyma, intercellular spaces are present.
 - Cells of parenchyma are generally isodiametric.
 - Sclerenchyma cells are usually dead without protoplasts.
 - Which of the following is not the function of parenchyma?
 - Photosynthesis
 - Buoyancy
 - Food storage
 - None of these
 - Identify the following figure and label A, B, C and D.



- Collenchyma. A= Thickened corners, B= Protoplasm, C= Vacuole, D= Cell wall
- Parenchyma. A= Thickened corners, B= Protoplasm, C= Vacuole, D= Cell wall

- c. Collenchyma. A= Thickened corners, B= Vacuole, C= Protoplasm, D= Cell wall
- d. Sclerenchyma. A= Thickened corners, B= Protoplasm, C= Vacuole, D= Cell wall
17. Collenchyma consists of cells with much thickened corners due to deposition of
- Cellulose, hemicellulose and chitin
 - Cellulose, lignin and pectin
 - Cellulose, hemicellulose and pectin
 - Cellulose, lignin and chitin
18. Regarding collenchyma, which statement is false?
- It occurs in layers below the epidermis in dicot plants.
 - It is found as a homogeneous layer or in patches.
 - Intercellular spaces are absent.
 - Cells are oval, spherical or polygonal and do not contain chloroplasts.
19. The tissue that provides mechanical support to the growing parts such as young stem and petiole of a leaf is
- Collenchyma
 - Parenchyma
 - Prosenchyma
 - Sclerenchyma
20. Analyze the below statements regarding sclerenchyma.
- It consists of long, narrow cells with thick and lignified cell walls having a few or numerous pits.
 - The cells are usually dead without protoplasts.
 - The fibres are thin walled, elongated and pointed.
- Of these,
- A & C are correct
 - B & C are correct
 - A & B are correct
 - All are correct
21. Odd one out
- Sieve tubes
 - Companion cells
 - Tracheids
 - Bast fibre
22. Two types of sclerenchyma are
- Tracheids and Sclereids
 - Sclereids and fibres
 - Vessels and fibres
 - Vessels and Tracheids
23. Major function of sclerenchyma is
- Mechanical support to organs
 - Role in photosynthesis
 - Secretion resins and gums
 - Protection of leaves
24. Select the true statements from the following.
- Parenchyma cells are living whereas sclerenchyma cells are dead.
 - Sclerenchyma cells retain their power of division whereas Parenchyma cells cannot divide.
 - Parenchyma cell walls are thin whereas the cell walls of sclerenchyma cells are highly lignified and hence thick.
 - Sclerenchyma cells contain cytoplasm whereas the Parenchyma cells are empty.
- A & C
 - B & D
 - Only A
 - A, B & C
25. The tissues made up of more than one type of cells and they work together as a unit are called

- Lateral meristems
 - Permanent tissues
 - Simple permanent tissues
 - Complex permanent tissues
26. Xylem is a conducting tissue which has 4 components such as
- Tracheids, vessels, xylem fibres and xylem parenchyma
 - Tracheids, sclerieds, xylem fibres and xylem parenchyma
 - Sclerieds, companion cells, xylem fibres and xylem parenchyma
 - Sclerieds, vessels, xylem fibres and companion cells
27. Identify the figure given below and label A & B.



- Sclerenchyma. A= Fibre, B= Sclerieds
 - Phloem. A= Fibre, B= Sieve tube elements
 - Xylem. A= Tracheid, B= Vessels
 - Sclerenchyma. A= Sclerieds, B= Fibre
28. Both xylem and phloem consist of
- Tracheids and fibres
 - Parenchyma and fibres
 - Sieve tube & fibres
 - Tracheids and parenchyma
29. Which of the following is *not* the similarity between xylem and phloem?
- Both have a tubular shape
 - Both are present in the roots, stems and leaves of the plant
 - Both perform bidirectional flow of materials
 - Both contain living cells and dead cells
30. Analyze the following statements.
- In xylem, of four types of cells, one is living and three are dead.
 - In phloem, of four types of cells, one is living and three are dead.
 - In xylem, of four types of cells, three are is living and one is dead.
 - In phloem, of four types of cells, three is living and one is dead.
- Of these,
- A & B are correct
 - A & B are wrong
 - A & D are correct
 - A & D are wrong
31. The xylem vessels are a characteristic feature of
- Bryophytes
 - Pteridophytes
 - Gymnosperms
 - Angiosperms

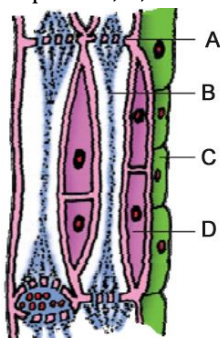
32. Which of the following statement correctly defines Endarch condition?

- It is a type of primary xylem seen in stem in which the protoxylem lies towards the centre (pith) and the metaxylem lies towards the periphery of the organ.
- It is a type of primary xylem seen in stem in which the protoxylem lies towards the periphery of the organ and the metaxylem lies towards the centre (pith).
- It is a type of primary xylem seen in root in which the protoxylem lies towards the centre (pith) and the metaxylem lies towards the periphery of the organ.
- It is a type of primary xylem seen in root in which the protoxylem lies towards the periphery of the organ and the metaxylem lies towards the centre (pith).

33. The components of phloem in gymnosperms are

- Sieve tube, companion cells, parenchyma and fibres
- Sieve tube, companion cells, albuminous cells and fibres
- Sieve cells, companion cells, albuminous cells and parenchyma
- Sieve cells, albuminous cells, parenchyma and fibres

34. Given below is the diagrammatic representation of phloem. Label the parts A, B, C and D.



- A= sieve tube element, B= sieve pore, C= phloem parenchyma, D= companion cell
- A= sieve pore, B= sieve tube element, C= phloem parenchyma, D= companion cell
- A= companion cell, B= sieve tube element, C= phloem parenchyma, D= sieve pore
- A= sieve tube element, B= phloem parenchyma, C= sieve pore, D= companion cell

35. Phloem of gymnosperms lacks

- Sieve tubes and companion cells
- Phloem parenchyma and companion cells
- Sieve tubes and phloem fibres
- Phloem parenchyma and phloem fibres

36. Companion cells are

- Specialized collenchymatous cells closely associated with sieve tube elements
- Specialized parenchymatous cells closely associated with bast fibres
- Specialized parenchymatous cells closely associated with sieve tube elements.
- Specialized sclerenchymatous cells closely associated with bast fibres

37. The function of phloem parenchyma is

- Storing of food material and other substances like resins, latex and mucilage.
- Conduction of food materials from leaves.
- Maintenance of the pressure gradient in sieve tubes.
- Mechanical support & protection to soft tissues.

38. Of the given statements regarding phloem fibres, one is wrong. Pick it out.

- They are made up of sclerenchymatous cells.
- They are generally present in primary phloem but are absent in the secondary phloem.
- They have quite thick cell wall.
- At maturity, they lose protoplasm and become dead.
- Phloem fibres of jute, flax and hemp are used commercially.

THE TISSUE SYSTEM

39. Cuticle is absent in

- Stem
- Root
- Leaf
- Both b and c

40. Regarding trichomes, which statement is incorrect?

- They are the epidermal hairs on stem.
- They may be secretory.
- They are unicellular, unbranched and soft.
- They help to prevent water loss due to transpiration.

41. In stomata, a few epidermal cells, near the guard cells become specialized in their shape and size. They are known as

- Mesophyll
- Subsidiary cells
- Bulliform cells
- Companion cells

42. Ground tissue includes

- All tissues except epidermis & vascular bundles
- Epidermis and cortex

c. All tissues internal to endodermis

d. All tissues external to endodermis

43. Lateral roots develop from primordia originated by the division of

- Pericycle cells lying opposite to protoxylem points
- Pericycle cells lying between two protoxylem points
- Endodermal cells lying between two protoxylem points
- Endodermal cells lying opposite to protoxylem points

44. Jute is mainly composed of

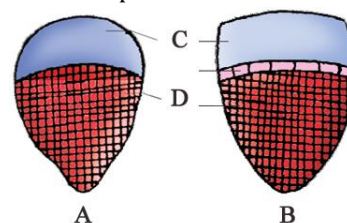
- Xylem
- Secondary bast fibre
- Phloem
- Cortex

45. The beneficial use of epidermal layer is

- Bast fibre
- Mesocarp

- c. Cotton fibre d. Jute
46. Vascular bundle having phloem at the centre encircled by xylem is known as
- a. Bicollateral b. Conjoint Collateral
c. Amphivasal d. Amphicribal
47. Bicollateral conjoint vascular bundles have
- a. Xylem & phloem, which are arranged in an alternate manner on different radii
b. Xylem & phloem, which are situated at the same radius and it has two groups of phloem along the two sides of xylem (inside and outside)
c. Xylem & phloem in same radius but it has only one group phloem outside the xylem
d. Phloem surrounds the xylem tissues
e. Xylem surrounds the phloem tissue
48. Amphivasal vascular bundle possess
- a. Xylem around phloem
b. Phloem around xylem
c. Phloem on both sides of xylem
d. Phloem towards centre and xylem towards periphery
49. Which of the following is not a characteristic feature of bryophytes?
- a. Dominant gametophytic generation
b. Filamentous rhizoids
c. Amphibious habitat
d. Vascular tissues
50. A bicollateral vascular bundle has the following arrangement of tissues.
- a. Outer phloem → Outer xylem → Middle cambium → Inner xylem → Inner phloem
b. Outer cambium → Outer phloem → Middle xylem → Inner phloem → Inner cambium

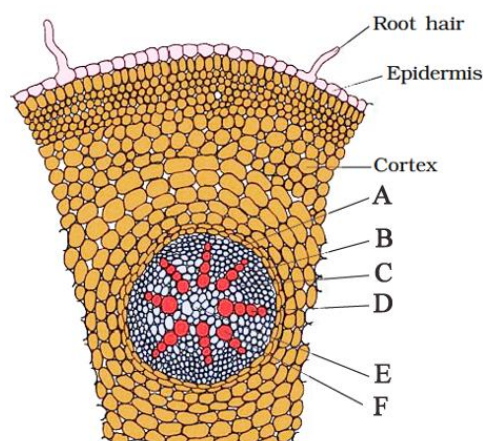
- c. Outer phloem → Outer cambium → Middle xylem → Inner cambium → Inner phloem
d. Outer xylem → Outer cambium → Middle phloem → Inner cambium → Inner xylem
51. In a vascular bundle, if xylem vessels develop in a centripetal fashion, the xylem is likely to be
- a. Mesarch b. Centrarch
c. Endarch d. Exarch
52. Meristematic tissue in vascular bundle is
- a. Phellem
b. Procambium
c. Interfascicular cambium
d. Fascicular cambium
53. Vascular bundle, in which two patches of phloem are present on both sides of xylem is
- a. Collateral b. Bicollateral
c. Concentric d. Radial
54. Given below is the representation of 2 types of vascular bundles. Select the option which is correctly labelled.



- a. A= Conjoint open, B= Conjoint closed, C= Phloem, D= Xylem
b. A= Conjoint open, B= Conjoint closed, C= Xylem, D= Phloem
c. A= Conjoint closed, B= Conjoint open, C= Phloem, D= Xylem
d. A= Conjoint closed, B= Conjoint open, C= Xylem, D= Phloem

ANATOMY OF DICOT & MONOCOT PLANTS

55. Casparian strip is found in
- a. Epidermis b. Pericycle
c. Endodermis d. Endothecium
56. All tissues on the inner side of the endodermis together constitute
- a. Stele b. Pith
c. Pericycle d. Conjunctive tissue
57. Vascular bundles in monocotyledons are considered closed because
- a. Xylem is surrounded all around by phloem
b. There are no vessels with perforations
c. A bundle sheath surrounds each bundle
d. There is no secondary growth
58. The stele is composed of
- a. Vascular bundle
b. Pith and vascular bundle
c. Cortex and endodermis
d. Pith and cortex
59. Given below is the figure showing the anatomy of monocot root. Label A to F correctly.

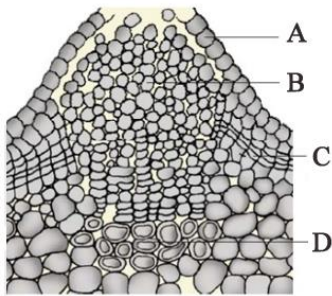


- a. A= endodermis, B= pericycle, C= protoxylem, D= phloem, E= pith, F= metaxylem
b. A= pericycle, B= endodermis, C= phloem, D= metaxylem, E= pith, F= protoxylem
c. A= endodermis, B= pericycle, C= phloem, D= protoxylem, E= pith, F= metaxylem
d. A= pericycle, B= endodermis, C= phloem, D= protoxylem, E= pith, F= metaxylem
60. Consider the following statements.

- I. In a dicot root, the vascular bundles are collateral and endarch.
 II. The innermost layer of cortex in a dicot root is endodermis.
 III. In a dicot root, the phloem masses are separated from the xylem by parenchymatous cells.
- I is true, II & III are false
 - II is true, I & III are false
 - I is false, II & III are true
 - II is false, I & III are false
 - III is true, I & II are false
61. Maximum number of vascular bundles are present in
 a. Monocot stem b. Monocot root
 c. Dicot stem d. Dicot root
62. The ring arrangement of vascular bundle is the characteristic feature of
 a. Monocot stem b. Dicot stem
 c. Monocot root d. Dicot root
63. In dicot stem, the cortex has multiple layers of cells arranged in between
 a. Hypodermis and endodermis
 b. Epidermis and cortical layers
 c. Epidermis and pericycle
 d. Hypodermis and pericycle
64. In dicot stem, which layer is known as starch sheath?
 a. Epidermis b. Hypodermis
 c. Cortical layers d. Endodermis
65. Which type of vascular bundles are found in monocot Stem?
 a. Collateral, open and endarch
 b. Radial, open and diarch
 c. Radial, open and mesarch
 d. Collateral, closed and endarch
66. Regarding monocot stem, choose the wrong statement.
 a. It has a sclerenchymatous hypodermis
 b. It has many scattered vascular bundles
 c. It has phloem parenchyma
 d. It has water-containing cavities within the vascular bundles
67. Palisade parenchyma and spongy parenchyma are the components of
 a. Mesophyll b. Epidermis
 c. Pericycle d. Vascular bundles
68. In dicotyledonous (dorsiventral) leaf, bundle sheath cells are seen surrounding the
 a. Epidermis b. Vascular bundles
 c. Mesophyll d. Stomata
69. In monocotyledonous (isobilateral) leaf, stomata are present on
 a. Lower surface of epidermis
 b. Upper surface of epidermis
 c. Both surfaces of the epidermis
 d. Mesophyll region
70. In grasses, certain adaxial epidermal cells along the veins modify themselves into large, empty, colourless cells. These are called
 a. Bulliform cells b. Guard cells
 c. Mesophyll cells d. Complementary cells

SECONDARY GROWTH

71. Interfascicular cambium is found
 a. Between two vascular bundles
 b. Between pith and vascular bundle
 c. In the vascular bundle
 d. Outside the bundle
72. Heartwood differs from sapwood in
 a. Presence of rays and fibres
 b. Absence of vessels and parenchyma
 c. Having dead and non-conducting elements
 d. Being susceptible to pests and pathogens
73. Tyloses an outgrowth from ray or axial parenchyma cell into the lumen of a vessel which partially or completely blocks the cavity are present in
 a. Periderm b. Heartwood
 c. Sapwood d. Secondary cortex
74. The study of determination of age of tree by counting annual rings is called
 a. Dendrology b. Chronology
 c. Dendrochronology d. Gerontology
75. Which one is known as early wood?
 a. Autumn wood b. Spring wood
 c. Heartwood d. Sapwood
76. Regarding activity of cambial ring, select the incorrect statement from the following
 a. The cambial ring becomes active and cut off new cells, towards the inner and outer sides.
 b. The cells cut off towards pith, mature into secondary phloem and the cells cut off towards periphery mature into secondary xylem.
 c. Cambium is more active on the inner side than on the outer.
 d. At some places, cambium forms a narrow band of parenchyma, which passes through the secondary xylem and the secondary phloem to form secondary medullary rays.
77. In a dicotyledonous stem, the sequence of tissues from outer to inner is
 a. Phellem – pericycle – endodermis – phloem
 b. Phellem – phloem – endodermis – pericycle
 c. Phellem – endodermis – pericycle – phloem
 d. Pericycle – phellem – endodermis – phloem
78. Phellogen, phellem, and phelloderm are collectively known as
 a. Periderm b. Pericycle
 c. Bark d. Late wood
79. A non-technical term that refers to all tissues such as periderm & secondary phloem exterior to the vascular cambium is
 a. Pericycle b. Cork
 c. Secondary cortex d. Bark
80. Choose the correct combination of labelling of a lenticel.



- a. A= complementary cells, B= epidermis, C= cork cambium, D= secondary cortex
- b. A= epidermis, B= complementary cells, C= secondary cortex D= cork cambium,
- c. A= epidermis, B= secondary cortex, C= cork cambium, D= complementary cells
- d. A= epidermis, B= complementary cells, C= cork cambium, D= secondary cortex

81. A nail is driven into the trunk of a 30 years old tree at a point 1 m above the soil level. The tree grows in height at the rate of 0.5 m a year. After three years, nail will be
 - a. 1 m above the soil level
 - b. 1.5 m above the soil level
 - c. 2 m above the soil level
 - d. 2.5 m above the soil level
82. Choose the wrong statement
 - a. In dicot root, vascular cambium is completely secondary in origin.
 - b. It originates from the tissue located just below the xylem bundles above the protoxylem.
 - c. Secondary growth also occurs in stems and roots of gymnosperms.
 - d. Secondary growth does not occur in monocotyledons.

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